

NAMSOS, Norway

WMO: 012900

Lat: 64.471N

Lon: 11.571E

Elev: 7

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.4	10.6	0.8	5.7	8.9	4.5	6.9	13.5	33.0	36.1	29.9	31.8	13.7	50	0.698

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	78.8	63.7	74.6	61.7	70.5	59.7	65.7	76.5	62.9	72.5	60.7	68.3	8.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.0	80.1	70.7	58.8	73.9	67.3	56.8	68.9	64.8	30.4	77.0	28.4	72.2	26.9	68.3	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.8	22.6	20.1	1.1	84.7	5.1	4.9	-2.6	88.2	-5.6	91.1	-8.5	93.9	-12.2	97.5		
(5)			0.5	68.1	5.0	3.4	-3.1	70.6	-6.1	72.5	-8.9	74.4	-12.6	76.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	28.1	29.8	32.7	39.3	47.8	53.0	59.2	57.6	51.6	42.2	35.2	31.8
	DBStd	12.72	8.35	8.38	7.60	5.84	7.45	5.55	7.19	5.06	5.44	5.43	6.73	8.05
	HDD50	3579	678	567	538	327	134	28	2	2	44	249	445	565
	HDD65	8281	1143	987	1003	771	533	360	213	236	402	708	895	1030
	CDD50	816	0	0	0	6	67	118	288	238	92	7	0	0
	CDD65	44	0	0	0	0	2	1	34	7	0	0	0	0
	CDH74	419	0	0	0	0	25	7	326	60	1	0	0	0
	CDH80	99	0	0	0	0	1	0	96	2	0	0	0	0
Wind	WSAvg	9.8	13.0	12.0	10.8	9.1	8.3	7.8	7.2	7.5	9.4	10.4	11.2	11.6
Precipitation	PrecAvg	50.9	5.4	4.4	4.7	2.9	2.5	3.3	3.4	3.4	5.1	5.2	4.7	5.4
	PrecMax	65.7	11.1	11.9	11.0	6.1	4.1	6.4	7.4	5.6	12.3	9.0	9.9	11.3
	PrecMin	34.1	0.2	0.7	1.2	0.8	1.0	0.4	1.2	0.8	1.3	1.5	0.6	2.2
	PrecStd	6.9	3.0	2.6	2.5	1.3	0.8	1.5	1.5	1.5	2.9	1.8	2.8	2.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.0	45.9	50.0	65.3	77.5	75.1	88.0	79.7	70.2	62.1	50.5	47.8
		MCWB	42.5	41.6	41.6	50.3	59.4	58.6	68.7	63.5	59.0	53.0	47.7	42.5
	2%	DB	42.8	44.2	46.0	57.9	72.1	70.3	82.4	75.7	66.1	55.6	48.1	45.5
		MCWB	40.0	40.1	39.8	46.1	57.4	57.9	65.5	62.1	57.3	49.7	45.7	41.4
	5%	DB	40.6	42.6	43.7	52.7	66.8	66.9	76.9	71.3	62.8	51.9	45.7	43.6
		MCWB	37.7	39.1	39.6	42.7	54.7	56.6	64.8	60.5	55.2	47.4	43.3	40.4
	10%	DB	38.7	40.5	42.1	48.5	61.1	63.7	72.7	67.4	60.0	49.4	43.1	41.5
		MCWB	36.0	37.4	38.7	41.4	52.4	55.2	62.2	58.7	53.6	46.1	40.7	38.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.6	43.1	43.9	51.5	61.2	62.4	70.7	65.7	60.2	54.3	48.5	45.2
		MCDB	44.7	45.3	46.4	64.6	74.0	72.3	83.9	76.0	67.6	60.1	49.5	46.0
	2%	WB	40.8	41.3	41.9	47.4	58.1	59.7	67.7	63.2	58.0	50.4	46.0	42.7
		MCDB	41.9	43.4	44.3	57.1	69.6	67.6	79.2	72.7	64.5	53.7	47.6	44.1
	5%	WB	38.0	39.7	40.5	44.1	55.4	57.3	65.3	61.2	56.1	48.3	43.3	40.7
		MCDB	39.9	42.1	43.3	50.2	65.8	64.6	75.9	69.1	61.5	51.4	44.9	42.8
	10%	WB	36.1	37.3	38.7	42.0	52.6	55.4	62.6	59.4	54.4	46.6	41.1	38.4
		MCDB	37.9	40.2	41.6	47.3	60.9	62.8	71.6	66.2	59.0	48.9	42.8	41.3
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.2	10.9	13.4	15.0	12.5	14.2	13.9	11.6	9.3	7.5	8.6
		MCDBR	9.2	10.2	11.4	21.2	24.8	20.1	23.9	21.1	16.9	12.9	9.3	10.4
		MCWBR	7.6	7.8	7.9	12.0	12.6	10.2	11.0	10.2	9.3	8.4	7.6	8.5
	5% WB	MCDBR	8.8	9.7	10.3	17.6	24.0	18.4	22.4	19.0	15.5	10.8	9.2	10.2
		MCWBR	7.5	8.3	8.0	10.7	12.7	9.7	10.7	10.0	9.1	7.6	8.1	9.0
Clear-Sky Solar Irradiance	taub	0.381	0.301	0.330	0.357	0.356	0.355	0.362	0.349	0.335	0.311	0.345	0.364	
	taud	2.600	2.178	2.239	2.334	2.422	2.459	2.466	2.493	2.516	2.388	2.513	2.507	
	Ebn at Noon	93	187	236	259	271	273	268	261	239	188	91	37	
	Edh at Noon	8	19	27	31	31	31	30	26	21	15	7	4	
All-Sky Solar Radiation	RadAvg	35	174	544	1081	1450	1548	1463	1124	639	275	62	14	
	RadStd	5	29	94	101	94	192	152	117	76	41	11	1	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)		+0.83	N/A	N/A	N/A	N/A	N/A	-257	-340	+121	N/A

Nomenclature: See separate page